

Plus Points in Implementation (Overall Evaluation Criteria)

1. Authentication:

- Implement robust user authentication protocols to ensure secure access.

2. Cost Estimation - Time and Space:

- Conduct a thorough analysis of time and space complexity in the system.
- Utilize efficient algorithms and data structures to optimize both time and space requirements.

3. Handling System Failure Cases:

- Implement fault-tolerant mechanisms to address system failures.
- Employ backup and recovery strategies for data integrity.
- Develop comprehensive error recovery procedures to minimize downtime.

4. Object-Oriented Programming Language (OOPS):

- Choose a robust OOPS language for structured and modular code.
- Leverage OOPS principles such as encapsulation, inheritance, and polymorphism for maintainability and extensibility.

5. Trade-offs in the System:

- Clearly define and document trade-offs made during system design.
- Evaluate and communicate the rationale behind architectural and design decisions.
- Consider trade-offs in terms of performance, scalability, and maintainability.

6. System Monitoring:

- Implement comprehensive monitoring tools to track system performance.
- Utilize real-time dashboards and logging mechanisms to promptly identify and address issues.

7. Caching:

- Integrate caching mechanisms to enhance system response times.
- Utilize caching for frequently accessed data to reduce database load.
- Implement cache eviction policies for optimal resource utilization.

8. Error and Exception Handling:

- Develop a robust error and exception handling framework.
- Provide meaningful error messages for effective debugging.
- Regularly review and update error-handling strategies based on system usage patterns.

Instructions:

1. Read and Understand the Problem Statement:

- Carefully read the problem statement provided. Understand the requirements, inputs, expected outputs, and any constraints mentioned.

2. Choose a Programming Language:

- Select a programming language you are comfortable with and that is suitable for solving the problem described in the case study.

3. Design Your Solution:

- Plan the overall structure of your solution. Consider the algorithms, data structures, and any potential optimizations needed.

4. Write the Code:

- Implement your solution in code. Follow best practices for coding standards, such as meaningful variable names, proper indentation, and comments where necessary.
- Break down the problem into smaller functions or modules to improve code readability and maintainability.

5. Test Your Code:

- Test your code thoroughly with different sets of input data, including edge cases and boundary conditions.
- Ensure that your code produces the expected outputs for all test cases.

7. Document Your Code :

- Consider adding documentation or comments to explain the logic and purpose of your code, especially for complex parts or algorithms.

8. Submit Your Solution:

- Once you're satisfied with your code and it meets all the requirements, submit your solution on GitHub and share the GitHub link.

9. Demonstration:

- Include a demonstration video showcasing key features of the ride-sharing platform.
- Alternatively, use screenshots to visually highlight the user interface and functionality.

Park and Ride: Smart Parking & Last-Mile Connectivity Solution

A **Park and Ride** system enhances urban mobility by integrating **parking reservations and last-mile transportation options**. This system allows commuters to **pre-book parking spaces, seamlessly transition to public transport**, and **reserve last-mile rides**, ensuring a **smooth, efficient, and hassle-free travel experience**. Below is a detailed breakdown of its functionalities:

I. Seamless Parking Booking

A **real-time parking reservation system** allows users to pre-book parking slots near metro stations, reducing last-minute parking stress.

Key Features

✓ Advance Booking System

- Users can **search for and book** parking spaces near metro stations.
- Reservation options for **hourly, daily, or monthly** parking.
- Users receive a **QR code or digital pass** for contactless entry.

✓ Smart Parking Spot Assignment

- The system dynamically **allocates parking spots** based on real-time availability.
- Ensures **optimal space utilization** by directing users to the nearest vacant slot.

✓ Flexible Cancellation & Modifications

- Users can **cancel or modify** their reservations before arrival.
- Refund policies based on **cancellation timing** (e.g., full refund if canceled 2 hours before).

✓ License Plate Recognition (LPR) & RFID Entry

- **Automated check-in/check-out** using **LPR cameras** and **RFID-enabled gates**.
- Reduces the need for manual intervention at entry points.

♦ Outcome:

- **No more last-minute parking struggles** – users have a reserved spot before arrival.
- **Minimizes congestion** at parking lots by optimizing space allocation.
- **Increases revenue for metro operators** through digital pre-booking and subscription plans.

II. Cab/Last-Mile Shuttle Integration

Seamlessly connects commuters to **cabs, e-rickshaws, or shuttle services** from metro stations, ensuring efficient last-mile connectivity.

Key Features

✓ Multi-Modal Booking

- Users can book a **cab, shuttle, or e-rickshaw** via the app upon exiting the metro station.
- **Auto-suggestions** based on commute history and user preferences.

✓ On-Demand & Scheduled Rides

- **Instant ride booking** for immediate departure.
- **Pre-scheduled rides** for a seamless transition from metro to final destination.

✓ Pooling & Ride-Sharing Options

- Shared **shuttle/cab pooling** to reduce costs.
- **AI-driven route optimization** to minimize detours.

✓ Integration with Public Transport

- Users can view **real-time bus & metro schedules** to plan their journey.
- Integration with **metro cards & digital wallets** for seamless payments.

♦ Outcome:

- **Eliminates last-mile travel challenges** by providing instant ride options.
- **Reduces traffic congestion** by promoting shared rides.
- **Enhances commuter experience** through **real-time ride tracking and reduced wait times**.

III. Availability Conflict Resolution

To prevent **overbooking and allocation errors**, the system ensures real-time **availability tracking and dynamic slot allocation**.

Key Features

✓ Live Slot Tracking

- The system **continuously updates parking availability** based on check-ins and check-outs.
- **Smart sensors & IoT-based tracking** help detect occupied and free spaces.

✓ Auto-Cancellation for No-Shows

- Reserved slots are **automatically released** if a user doesn't check in within a grace period.
- Ensures **fair allocation** to other users in need of parking.

✓ Dynamic Slot Reassignment

- If a user arrives earlier than expected, the system **automatically reallocates** a vacant slot.
- **Prevents double booking** through instant updates across all users.

♦ Outcome:

- **Reduces waiting time** at parking entrances.
- **Prevents disputes over reservations** by ensuring fair and accurate slot allocation.
- **Maximizes parking lot efficiency** through real-time availability tracking.

IV. Dynamic Pricing

To **balance demand and optimize revenue**, parking fees and ride fares are adjusted dynamically based on **traffic flow, peak hours, and occupancy levels**.

Key Features

✓ Demand-Based Pricing

- Prices are **higher during peak hours** to **control demand**.
- **Discounted rates** during **off-peak hours** encourage more users.

✓ Subscription Plans & Loyalty Rewards

- Monthly **subscriptions** for frequent commuters.
- **Reward points** for regular users, redeemable for discounts on parking and last-mile rides.

✓ AI-Based Surge Pricing

- Prices **automatically adjust** based on **real-time traffic, weather, and event schedules**.
- Surge pricing ensures **availability of parking spots** for those who need them most.

♦ Outcome:

- **Encourages users to travel during off-peak hours**, reducing congestion.
- **Increases revenue** for metro operators while maintaining fair pricing.
- **Enhances parking availability** by discouraging unnecessary long-term occupancy.

V. Offline Mode Handling

Ensures seamless operation **even in low-connectivity zones** such as metro stations and underground parking areas.

Key Features

✓ Offline Booking Access

- Users can **view existing reservations** without an internet connection.
- QR codes or **digital passes are accessible** offline for easy check-in.

✓ Stored Navigation Routes

- Users can **preload maps** of the parking area & metro station.
- **Voice-guided navigation** to guide users to their reserved spot even without network access.

✓ Auto-Sync on Reconnection

- Once the device regains connectivity, **all offline actions sync automatically**.
- Users receive pending **notifications and ride updates**.

◆ Outcome:

- **Ensures uninterrupted access to parking and ride details** even in metro basements.
- **Prevents check-in delays** due to network failures.
- **Enhances user experience** with a **smooth, hassle-free transition** from parking to public transport.